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Wireless Fast Phone Charger

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Introduction

A wireless fast phone charger is a power-transfer device that supplies electrical energy to a smartphone without requiring a direct wired connection. The device transfers energy when a compatible smartphone is placed directly on its upper surface. The charger described in this paper is circular, black in color, and constructed of rigid polymer plastic. It measures approximately 10 cm in diameter and 0.8 cm in height. The charger consists of four primary parts: the charging surface, the indicator light ring, the protective casing, and the power input port.

Discussion

Figure 1 shows the wireless fast phone charger and its primary components.

Charging Surface

The charging surface is a circular support platform that stabilizes a smartphone during electromagnetic energy transfer. The surface measures approximately 10 cm in diameter with a thickness of 0.8 cm. It is constructed of smooth matte plastic with a slightly textured finish that increases friction between the phone and the charger. The flat circular design ensures that the smartphone remains centered during operation. Proper alignment allows consistent electromagnetic induction between the charger and the device battery system.

Indicator Light Ring

The indicator light ring is a visual signaling component that communicates charging status to the user. The ring follows the outer perimeter of the charging surface and has an approximate thickness of 0.2 cm. It is embedded within the upper casing and constructed of translucent plastic. When the charger is connected to power, the ring emits LED light. The color of the light

changes depending on operational status, providing observable feedback regarding charging activity.

Protective Casing

The protective casing is a structural enclosure that houses and supports the internal electrical components of the charger. The casing is circular with a diameter of approximately 9.5 cm and a total height of 0.8 cm. It is constructed of rigid black polymer plastic with a smooth exterior finish. The casing protects the internal induction coil and circuitry while maintaining the structural integrity of the device. The underside of the casing contains small rubber pads that reduce slipping during operation.

Power Input Port

The power input port is an electrical connection interface that allows the charger to receive energy from an external power source. The port is located on the side of the casing and accommodates a USB-C connector. It measures approximately 8 mm in width. The port is reinforced within the plastic housing to withstand repeated cable insertion and removal. When connected to a power adapter, the port supplies electrical current to the internal induction system.

Conclusion

A wireless fast phone charger is a power-transfer device that supplies electrical energy to a smartphone without requiring a direct wired connection. The charger consists of four observable components: the charging surface, which supports and aligns the smartphone; the indicator light ring, which provides operational feedback; the protective casing, which houses internal components; and the power input port, which receives external electrical power. These parts function together to create a compact and effective mechanism that supports modern mobile computing devices.



Figure 1: Wireless Fast Phone Charger, showing its parts.